

Mathematics for Computing

Assignment 1 : Set Theory

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Assignment No.1

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- General form of inclusion - exclusion principle
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Introduction

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- Example of set: $\{a, e, i, o, u\}$
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- Example of set: $\{1,7,19,20,14\}$
- Example of set: $\{a,e,i,o,u\}$
- Example of a set: Empty set (no items): $\{\}$ or $\emptyset$
- Example of **not** a set: Set of all tall people in this room
- Example of **not** a set: The collection of the best football players in the world.

Set Builder Form

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Set Builder Form:

$$Q_1 = \{ (x, y) \mid x, y \in \mathbb{R} \text{ and } x > 0 \text{ and } y > 0 \}$$

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$$S = \{x \mid x^2 < 50 \forall x \in \mathbb{Z}\}$$

Sets used in A.I. and M.L. Domain

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- **Test Set**

$\mathbf{D}_{Test} = \{(x_i, y_i) \in D \mid i \in \mathbf{N} \text{ and } i \notin \mathbf{I}_{Train}\}$ where $\mathbf{I}_{Train}$ contains the specific training indices.

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